University of Glasgow

TODO DATE AND TIME
Duration: 2 hours
Additional time: 30 minutes
Timed exam – fixed start time

DEGREE of MSc

Big Data: Systems, Prog & Mgt M
COMPSCI 5088

(Answer all questions)

This examination paper is an open book, online assessment and is worth a total of 75 marks.

1. **(a)** **Context**: Consider the following scenario. You are a big data engineer working at a satellite imaging company, who are contracted to develop an alerting system for natural disasters based on the images captured by the satellites. When active, a satellite sends a high-resolution image to a base station on the ground each second, which is then streamed to the company headquarters to be processed. Each image is around 10 MB in size. The company owns 100 satellites with the goal of doubling that in the next 5 years. Satellites are set in fixed orbital paths and are tasked with taking pictures from specific regions on earth. When not over a target region the satellite is inactive (to save power). Typically, around 20% of the satellites are active at any one time, although this may increase to 50% during busy periods of the year (e.g. US Hurricane season).

   Question: Processing the images produced by these satellites for disaster alerting constitutes a big data task. List the dimensions/properties of the task that makes it a big data problem and discuss why.

   Guidance: The question is worth 6 marks. There are 3 marks for correctly identifying the dimensions/properties and 3 marks for the associated discussion (1 mark per correct point made).

   [6]

   Solution:

   - Volume [1 Mark]: Up-to 500MB of images arriving each second constitutes a large volume of data to be processed [1 Mark]
   - Velocity [1 Mark]: The use-case for this data processing is real-time alerting and hence there are latency requirements for processing. [1 Mark].
   - Variability [1 Mark]: The amount of data needing processed varies over time based on real-world events [1 Mark].

   Marks could also be awarded for Veracity if a reasonable argument is made about error correction or data cleaning.

   (b) **Context**: The company is approached by a large hardware manufacturer offering a single mainframe computer to perform the processing of the satellite data.

   Question: Discuss why this is not a good option given their requirements.

   Guidance: The question is worth 8 marks. You are expected to make 4 distinct points in your discussion, 2 marks per point.

   [8]

   Solution:

   We expect the student to provide a discussion on the limitations of vertical scaling solutions vs. horizontal scaling solutions. Possible valid points include:

   - Mainframes have a maximum limit to scaling (both in compute and storage) [2 Marks]
   - Mainframes are more costly than horizontal scaled solutions [2 Marks]

- The company plans to scale-up their operations, meaning that a mainframe solution may become un-fit for purpose soon after installation [2 Marks]
- The mainframe acts as a single point of failure [2 Marks]

Other valid points are acceptable, 2 marks per point

(c) **Context**: The company instead decides to develop a horizontally scalable solution to this problem comprised of three components, each of which can be individually parallelised. These components are: 1) Image splitting into smaller regions (converts each image into 24 smaller regional images); 2) Region image variant generation (applies 4 different filters, e.g. grayscale, to each regional image, creating 4 variants of that image); and 3) disaster type classification on each regional image variant. If any of the regional image variants are classified as showing a natural disaster, then an alert is generated.

Question: Discuss what the main challenges are that the company will face when developing this solution.

Guidance: The question is worth 8 marks. You are expected to discuss 4 challenges. There is 1 mark for discussing a relevant parallelism concept and 1 mark for describing why it is a relevant challenge for this proposed system.

[8]

Solution: It is expected for the students to discuss work distribution challenges from W2-01. Expected points include:

- Concept: Data Splitting and Load Balancing [1 Mark]. Linking: the input images will need to be divided up amongst different machines such that all machines receive a roughly equal load. [1 Mark]
- Concept: Communication and Intermediate Storage Overheads [1 Mark]. Linking: The original images and the variants that are generated will need to transfer over the network and possible stored, causing scaling read/write overheads [1 Mark].
- Concept: Data Duplication [1 Mark]: In this solution an image is split into 24 parts, then duplicated 4 times, this is going to increase the amount of data needing processed in the later stages of the pipeline [1 Mark].
- Concept: Fault Tolerance [1 Mark]: As we are scaling up the pipeline, there is a need for fault-tolerance such that images or portions of images are not lost [1 Mark].

Marks can be awarded for other relevant concepts

2. **(a)** **Context**: A news aggregator company is experimenting with automatic summarization of news articles. They pay major news outlets to receive copies of news articles that those outlets publish and then algorithmically convert multiple news articles on a particular topic into a short summary for customers. Each article is stored in a separate file. The summarization process has four main stages: 1) content extraction from the news articles (converts the article into paragraphs); 2) topic identification, where each paragraph is assigned one or more topics (e.g. War in Ukraine, US Election, etc.); 3) paragraph grouping by topic; and 4) summarization of a group of paragraphs on each topic (converts multiple paragraphs into a single summary).

Question: The news aggregator company has decided to implement their automatic summarizer in MapReduce using Hadoop. Describe how you would implement this as a Hadoop program and why.

Guidance: The question is worth 9 marks. We are looking for you to describe how you convert the above stages into MapReduce operations. You should state what the type of each operation is and what that operation does in the context of this solution, along with its inputs and outputs (worth 6 marks). After describing your operations, you should then include a short paragraph (2-3 sentences) summarizing why you chose this design (worth 3 marks).

[9]

Solution: The correct solution is:

- Map 1: Content extraction + topic identification [1 Mark]. Input: <articleid, article>. [1 Mark] Output: <topic,paragraph> [1 Mark]
- Reduce 1: Performs summarization [1 Mark]. Input <topic,List<paragraph>> [1 Mark], Output summary [1 Mark].

Why: Content extraction and topic identification can be executed per-article, and so can be aggregated into a single step [1 Mark]. We want to group our paragraphs by topic, and the reducer will do that automatically if we set the keys provided as output of the map function as the topics [1 Mark]. Our summarization algorithm is then a reducer [1 Mark].

(b) Question: The company implements the above system in Java via Hadoop and then profiles it to determine where the main overhead costs are when processing. Describe what you expect them to find and why.

Guidance: This question is worth 8 marks. We expect you to list four main observations, where one mark is available for the observation and one mark is available for why. We do not expect you to discuss the cost of the operations themselves (since you won't know the relative cost of content extraction, topic identification or summarization), but rather aspects of Hadoop that will result in efficiency overheads.

[8]

Solution:

We are expecting them to discuss the disadvantages of Hadoop applications in general and this implementation in particular. Expected points include:

- Data loading will be slow [1 Mark] because each news article is stored in a separate file and the HDFS does not like small files. [1 Mark]

- There will be some cost performing the intermediate sort [1 Mark] of the data based on the output of the topic identification [1 Mark]
- The there will be significant network transfer latencies [1 Mark] as map outputs are sent across the network as it groups by key [1 Mark].
- There will be disk write costs [1 Mark] as paragraphs are persisted to disk after the map sort [1 Mark].

(c) **Context**: Given the overhead costs observed, the company decides that they should test a Java implementation in Apache Spark.

Question: Discuss what the primary advantages that the Spark implementation of this system would have over a Hadoop implementation.

Guidance: This question is worth 8 marks. 4 points are expected, where one mark is available for what the advantage is, and 1 mark is available for why this is better than what Hadoop provides.

[8]

Solution: We expect the students to describe the main advantages that Spark has over Hadoop, but avoiding any that are not relevant for this application. Valid points include:

- Initial data read costs will be lower [1 Mark] as Spark does not rely on the HDFS [1 Mark]
- Spark does not write intermediate data to disk [1 Mark], so those costs will be avoided [1 Mark]
- Transformation overheads will generally be lower [1 Mark] due to better serialization [1 Mark]
- Individual operations can be defined as separate transformations rather than needing to manually group them [1 Mark], as the structure of a Spark program is not rigid like MapReduce [1 Mark].

A point discussing more efficient scheduling or failure recovery would also be valid answers here.

Invalid points would be discussions on chaining maps/reduces, discussions on streaming or language support.

3. **(a)** **Context**: A large shipping company is in the process of upgrading the storage infrastructure that their company uses to store data produced by the ships in their fleet. The company needs to store two types of data: 1) audio logs of communication between its ships and third parties (coast guards, docking facilities, etc.) that are relatively large files and 2) sensor data produced by the ships themselves (GPS coordinate data, engine temperature, etc.) that comprise many small files.

Question: The company currently uses an NFS backed by RAID6 storage, which is replicated to an off-site back-up at the end of each week and is considering moving to an HDFS-based solution. What are the advantages and disadvantages of such a move?

Guidance: This question is worth 8 marks. 4 distinct points are expected. For each point, one mark is available for what difference between the two solutions would be and a further mark is available for what impact this would have. You should connect these differences to Big Data aspects like resiliency, scalability, etc.

[8]

Solution:

We expect the student to compare and contrast the NFS+RAID solution with HDFS. Valid points for discussion include:

- Moving to an HDFS solution would provide replication of files on-site [1 Mark], enabling better throughput due to more disk and network capacity [1 Mark].
- Having a cluster of machines [1 Mark] protects the company against hardware failure in the NFS server (a single point of failure) [1 Mark].
- On the other hand, an HDFS is more complex to manage and requires orchestration [1 Mark] as multiple servers running data nodes are needed [1 Mark].
- An HDFS solution would also be more expensive [1 Mark], as more redundant hardware would be needed [1 Mark].
- HDFS Storage policies can be used to support intelligent selection of storage devices for files [1 Mark], which is not supported by the NFS [1 Mark].
- Data access speeds will become faster for audio log retrieval, but slower for sensor data [1 Mark], as the HDFS is optimised for large file IO [1 Mark].

Other answers are acceptable under the same paired mark format.

(b) **Context**: One of the engineers at the company is worried about the Name Node as a single point of failure and instead suggested using a decentralized solution.

Question: Summarize the primary difference between a distributed and decentralized storage solution.

Guidance: This question is worth 3 marks. You should define both distributed and decentralized storage (2 marks) and then contrast them (1 mark).

[3]

Solution: In a distributed solution, storage is spread out amongst many physical machines, and usually replicates data across multiple of those machines [1 Mark]. However, in most distributed storage solutions (e.g. the HDFS), management of that storage process is controlled by a centralised service or master [1 Mark]. Decentralized solutions are a subset of distributed solutions which lack this centralized service/master [1 Mark].

(c) **Context**: One of the foundations of a decentralized solution like Cassandra is that it uses hashing schemes to enable fast indexing of its available storage.

Question: Describe consistent hashing and order-preserving hashing, and then discuss how they are combined in Cassandra.

Guidance: This question is worth 6 marks. 4 marks are available for explaining how the two hashing schemes function, and 2 marks are available for describing how they can be combined.

[6]

Solution: Both schemes are referred to as hashing schemes, but operate at different levels. Consistent hashing will partition the ID space across existing nodes, [1 mark] and will assign an ID X to the first node whose ID is larger than (or equal) to X. [1 mark] This applies regardless of how 'X' was computed. An order-preserving hash function can then be used to map an input keys Y to an ID A on the ID space in an order-preserving manner [1 mark] (i.e., guaranteeing that any key Z>Y will be mapped to an ID B>A), then consistent hashing would be used to map A to a node. [1 mark]. Combination discussion is worth [2 marks].

4. (a) Context: In Hadoop 2.0, Yet Another Resource Negotiator (YARN) was introduced.

Question: Discuss why a YARN-based Hadoop environment is superior to a traditional Hadoop configuration.

Guidance: This question is worth 3 marks.

[3]

Solution: We expect the student to list the advantages that YARN brings in terms of orchestration, specifically:

- YARN enables better use of available resources [1 Mark] as it can handle variable size resource requests rather than using fixed task slots [1 Mark].
- YARN Enables multiple types of application to work concurrently within the Hadoop ecosystem (e.g. Spark and HBase) [1 Mark].

(b) Context: In the original YARN paper it refers to the place where work is done as a 'container'. However, this is not the common meaning of containers that are used today.

Question: What are the differences between a YARN container and containers that are common today?

Guidance: The question is worth 8 marks. 4 differences are expected, 2 marks per difference. You need to specify how the two types of containers differ to receive the marks.

[8]

Solution: The student should compare YARN containers with Microservice/Docker containers. Expected differences highlighted include:

- In YARN a container represents a unit of work to be done, which will be physicalised as an OS process (typically a JVM process). In contrast, a microservice container is not only comprised of the code to be run, but also the operating system context (such as files and installed libraries).

- YARN containers are executed by a Node Manager process on each physical machine, while microservice containers are executed by a OS-level virtualization engine, such as Docker or Kubernetes.
- YARN only supports Hadoop ecosystem processes, while microservice containers can support any application.
- YARN containers share the OS environment of the host machine, while microservice containers are sandboxed with their own drivers and libraries.
- A YARN container can monitor the entire host machine, while microservice container can only see (and use) the resources that they are allocated.